

Interoperability of Information Systems in the Ecuadorian Public Sector

ARIAS, CARRERA, MERINO CALDERON, SUQUILLO

Software Development Applied to Interculturality

Department of Computer Science



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Introduction: The Path Toward Interoperability

Fragmented State

Historically, the citizen acted as the integrator of a siloed system:

- Absolute dependence on physical media and institutional stamps.
- High transactional costs (time and travel).
- Progressive loss of trust in public management.

Optimization Law (2018)

A legal paradigm shift through **Article 9**:

- Prohibition of requesting requirements that the State already possesses.
- Mandatory interconnection through the BSG.
- **Goal:** 77 % reduction in administrative in-person attendance.

Current Context

Interoperability ceases to be a technical option and becomes a legal mandate that guarantees the right to administrative agility.



Research Design

Qualitative approach with descriptive-exploratory scope. Systematic review of technical and legal regulations between 2018 and 2025.

Digital Maturity Scale (Based on ECLAC)

- **Informative/Interactive:** Web presence and form downloads.
- **Transactional:** Digital management with persistence of external requirements.
- **Transformative:** Full interoperability through real-time web service consumption ("Once-Only" Principle).



Evolution of the Regulatory Framework (2018–2025)

Chronology of Digital Transformation

- **Agreement 005-2018:** Establishes the Interoperability Framework and its technical, semantic, organizational, and legal layers.
- **Administrative Procedures Law 2018:** Introduces administrative speed and the prohibition of redundant requirements.
- **LOPDP 2021:** Regulates the secure exchange of information under personal data protection standards.
- **Digital Transformation Law 2023:** Legally validates electronic records and digital signatures across the state apparatus.

A legal ecosystem designed for the elimination of physical media.



Model of Centralized Exchange

Infrastructure based on an **Enterprise Service Bus (ESB)** acting as a logical orchestrator between provider nodes (servers) and consumers (entities).

Pillar 1: Communication

Use of standards for technical interoperability:

- **SOAP/XML:** For robust transactionality in legacy systems.
- **REST/JSON:** Agile integration for the *gob.ec* portal and mobile services.

Pillar 2: Security

Guarantee of data integrity and confidentiality:

- **Identity:** Access tokens and X.509 digital certificates.
- **Traceability:** Audit log for each request made.



Success Cases: Provision of Critical Information

Strategic Institutions

The Civil Registry and the SRI act as the provider nodes with the highest consumption rate in the Governmental Service Bus (BSG).

Civil Registry (Identity)

Primary data validation:

- Real-time identity and biometric verification.
- Civil status and death records query.
- **Effect:** Total replacement of the physical ID card copy.

SRI (Tax Management)

Tax facilitation and control:

- Instant RUC and solvency validation.
- **Efficiency:** Pre-filling of tax returns through data cross-referencing.
- Integration with public banking for credit history.

“Once-Only” Principle: The State already knows the citizen.



Maturity Analysis by Sector

Sector	Maturity Status	Lead Entity
Fiscal / Tax	Transformative	SRI
Identity and Civil	Transformative	Civil Registry
Higher Education	Transactional	SENESCYT
Public Health	Transactional	MSP / IESS
Municipal Management	Interactive / Initial	Municipal GADs

Critical Finding

Technological asymmetry: The Executive Branch leads adoption, while GADs present a significant infrastructure gap.



Challenges: The Organizational and Human Factor

Culture of the “Copy”

Resistance from officials due to fear of Comptroller audits. Demand for paper as “backup.”

Semantic Barrier

Lack of a unified data ontology. Concepts like “address” vary between institutions.

The LOPDP Challenge

Legal complexity in justifying data consumption, which slows down the integration of new services.



Conclusions and Future of Interoperability

- **GAD Standardization:** It is urgent to reduce the digital divide in small municipalities through SOA architectures.
- **Digital Trust:** Evolving from a culture of stamps to the traceability of electronic records.
- **Data Sovereignty:** Implementing models where the citizen authorizes the flow of their data in real time.

Interoperability is the foundation of the modern State. Its success depends on harmonizing **technical efficiency** with **legal security**.



Questions?

Arias, Carrera, Merino Calderón, Suquillo
Thank You Very Much

